

TSA Meeting — Slide 10 Study Note

Ch 6: SAMBP-87T — Transformer Differential with SPRT

Contribution C1 · Chapter 3 · SAMBP Framework
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Slide Overview and Narrative

What This Slide Shows

Slide 10 presents the **SAMBP-87T adaptive transformer differential relay**, the second component of Contribution C1. The narrative has three parts:

The problem: Transformer energisation produces inrush currents with large 2nd-harmonic content. Conventional relays block trips when $I_{2f}/I_{1f} > 15\%$ (2nd-harmonic criterion). But Type-4 IBR inverters inject 2nd-harmonic current continuously into the network, making the 2nd-harmonic ratio elevated even during genuine internal faults — causing false blocking and missed trips (8.7% false positive rate for the conventional relay).

The solution: Replace the fixed 2nd-harmonic threshold with a **Wald Sequential Probability Ratio Test (SPRT)** that accumulates log-likelihood evidence sample-by-sample, deciding “inrush” or “fault” as soon as the accumulated evidence crosses a statistical boundary. SAMBP further updates the H_1 (fault) likelihood model in real time using the IBR harmonic model derived from the EKF digital twin.

The result: Inrush rejection rises from 94.1% to 99.6%; IBR harmonic false positives drop from 8.7% to 0.9%; mean decision time halves from 14 to 7 samples.

Slide Key Results

Scenario	2H Block	SPRT	Change
Inrush rejection	94.1%	99.6%	+5.5 pp
Internal fault TPR	96.8%	99.1%	+2.3 pp
IBR harmonic FP	8.7%	0.9%	−90% relative
Mean decision (samples)	14	7	halved

Key observation from slide: SPRT halves the detection window *and* reduces IBR-induced false positives by 90% compared to 2nd-harmonic blocking. These are independent improvements — speed and selectivity both improve simultaneously.

Engineering Challenge: Why 2nd-Harmonic Blocking Fails for IBR

The Core Problem: IBR Harmonic Injection

Conventional transformer differential relays use the 2nd-harmonic ratio to distinguish inrush from fault:

$$\text{Block if: } \frac{I_{2f}}{I_{1f}} > h_{\text{th}} \approx 15\%$$

This works because inrush currents are rich in 2nd harmonic while genuine internal faults produce predominantly fundamental-frequency differential current.

Why IBR breaks this: Type-4 grid-following converters (full-power converters) inject harmonic currents into the network as a side effect of their PWM switching. The 2nd-harmonic injection level during normal IBR operation can reach $I_{2f}/I_{1f} \approx 8\text{--}25\%$ — overlapping with the inrush signature range. When an internal fault occurs in an IBR-dominated network:

1. The fault-induced differential current is small (k_{ibr} -limited to 1.1–2.0 pu)
2. The IBR continues to inject harmonic current through the fault
3. The combined 2nd-harmonic ratio I_{2f}/I_{1f} stays above h_{th}
4. The conventional relay blocks — *missing a genuine internal fault*

This is the “IBR harmonic FP” problem: the relay sees a high 2nd-harmonic ratio, assumes inrush, and incorrectly blocks. With conventional 2H blocking: 8.7% false positive rate (blocking a real fault). With SPRT: 0.9%.

Five-Parameter Zone Model for 87T (TR-03)

Before the SPRT, the SAMBP framework decomposes the differential current into identifiable components using a five-parameter zone model:

$$\hat{i}_{\text{diff}}(t; \boldsymbol{\theta}^T) = I_f \sin(\omega t + \varphi) + k_2 I_f \sin(2\omega t + \varphi) + k_5 I_f \sin(5\omega t + \varphi) + \varepsilon_{\text{CT}} I_f \text{sgn}(\sin \omega t)$$

with $\boldsymbol{\theta}^T = [I_f, \varphi, k_2, k_5, \varepsilon_{\text{CT}}]^T$. The internal-fault indicator is:

$$f_{\text{int}} = 1 - \frac{k_2/\delta_2 + k_5/\delta_5 + \varepsilon_{\text{CT}}/\delta_\varepsilon}{1/\delta_2 + 1/\delta_5 + 1/\delta_\varepsilon}$$

where $\delta_2 = 0.15$, $\delta_5 = 0.35$, $\delta_\varepsilon = 0.10$ are blocking thresholds. This model correctly classifies all 7 canonical events (TR-03 Table 3). The SPRT then replaces the fixed h_{th} threshold in the blocking logic with a statistically optimal sequential test.

Wald Sequential Probability Ratio Test (SPRT)

SPRT Log-Likelihood Ratio Accumulation

Let $\{x_i\}_{i=1}^k$ be sequential observations of the normalised 2nd-harmonic ratio. The two hypotheses are:

$$H_0 : x_i \sim f_0(x_i) \quad (\text{inrush or IBR harmonic — no fault})$$

$$H_1 : x_i \sim f_1(x_i) \quad (\text{internal fault})$$

The SPRT accumulates the log-likelihood ratio:

$$\Lambda_k = \sum_{i=1}^k \ln \frac{p(x_i | H_1)}{p(x_i | H_0)}$$

and decides at the first sample n when the accumulated evidence crosses a boundary:

$$\Lambda_n \geq A = \ln \frac{1 - \beta}{\alpha} \Rightarrow \textbf{Trip} \quad (\text{fault confirmed})$$

$$\Lambda_n \leq B = \ln \frac{\beta}{1 - \alpha} \Rightarrow \textbf{Restrain} \quad (\text{inrush confirmed})$$

where α = false trip probability and β = false restrain (miss) probability. For $\alpha = \beta = 0.01$: $A = \ln(99) \approx 4.60$ and $B = \ln(0.0101) \approx -4.60$.

The IEC 60255 Connection

The thresholds A and B are set to achieve specific α and β rates directly from the Wald boundary equations — this is what the slide means by “Thresholds A , B set to achieve $\alpha = 0.01$, $\beta = 0.01$ (IEC 60255)”. IEC 60255-151 specifies maximum false-trip and miss rates for protection relays; the SPRT boundary equations translate those requirements directly into computable thresholds, without ad-hoc tuning.

Contrast with conventional blocking: The 2H blocking threshold $h_{th} = 15\%$ has no direct connection to a false-trip probability. Setting it higher reduces false trips on faults but increases missed inrush blocks; setting it lower does the reverse. The SPRT threshold gives explicit control of *both* error rates simultaneously.

Harmonic Envelope Likelihood Models

Observable: Normalised 2nd-Harmonic Ratio

The observable quantity at each sample time t_k is:

$$\rho_k = \frac{I_{2f}(t_k)}{I_{1f}(t_k)}$$

computed from one-cycle DFT windows. ρ_k has different statistical distributions under inrush vs internal fault:

Hypothesis	Distribution	Physical meaning
H_0 (inrush)	$\rho_k \sim \mathcal{N}(\mu_0 = 0.35, \sigma_0 = 0.08)$	High 2H ratio, large variance
H_1 (fault)	$\rho_k \sim \mathcal{N}(\mu_1 = 0.06, \sigma_1 = 0.03)$	Low 2H ratio, small variance

The Gaussian parameters (μ_0, σ_0) were estimated from Monte Carlo simulation of 1000 inrush events across 20 transformer models. (μ_1, σ_1) reflect IBR fault currents with minimal harmonic content.

LLR Increment for Gaussian Models

With Gaussian distributions, the LLR increment at each sample is:

$$\ln \frac{f_1(\rho_k)}{f_0(\rho_k)} = \underbrace{\frac{(\rho_k - \mu_0)^2}{2\sigma_0^2}}_{\text{inrush penalty}} - \underbrace{\frac{(\rho_k - \mu_1)^2}{2\sigma_1^2}}_{\text{fault penalty}} + \ln \frac{\sigma_0}{\sigma_1}$$

Intuition:

- If $\rho_k \approx 0.35$ (high 2H, looks like inrush): the fault penalty is large, the inrush penalty is small $\Rightarrow$ LLR *decreases* $\Rightarrow \Lambda_k$ moves toward B (restrain)
- If $\rho_k \approx 0.06$ (low 2H, looks like fault): the inrush penalty is large $\Rightarrow$ LLR *increases* $\Rightarrow \Lambda_k$ moves toward A (trip)
- Ambiguous values ($\rho_k \approx 0.20$): LLR $\approx 0 \Rightarrow$ decision deferred; SPRT waits for more evidence

IBR-Adaptive Update of the H_1 Likelihood

The SAMBP enhancement over classical SPRT is the adaptive H_1 model: SAMBP updates μ_1 in real time using the IBR harmonic injection model from the EKF digital twin:

$$\mu_1(k_{\text{ibr}}) = \mu_1^{(0)} + \Delta\mu_1(k_{\text{ibr}})$$

where $\Delta\mu_1(k_{\text{ibr}})$ accounts for the residual IBR harmonic component expected in the fault current at the current fleet composition. This shifts the H_1 distribution rightward as k_{ibr} increases, preventing the SPRT from treating IBR-contaminated fault signatures as inrush.

Expected Decision Time (Wald's Identity)

Wald's Identity: Expected Number of Samples to Decision

Under H_1 (fault present), the expected number of samples before the SPRT crosses threshold A is:

$$\mathbb{E}[N \mid H_1] \approx \frac{A}{\mathbb{E}\left[\ln \frac{f_1(\rho)}{f_0(\rho)} \mid H_1\right]}$$

Using $\mu_1 = 0.06$, $\mu_0 = 0.35$, $\sigma_0 = 0.08$, $\sigma_1 = 0.03$:

$$\mathbb{E}\left[\ln \frac{f_1}{f_0} \mid H_1\right] = 2.73 \text{ per sample}$$

$$\mathbb{E}[N \mid H_1] \approx \frac{4.60}{2.73} \approx 1.7 \text{ cycles} \approx 34 \text{ ms at } 50 \text{ Hz}$$

The slide reports **mean decision = 7 samples** in the SPRT case vs 14 samples for 2H blocking. The improvement matches the theoretical Wald bound: the SPRT is statistically optimal in the sense that *no other sequential test requires fewer samples on average* to achieve the same (α, β) guarantees.

Why 7 Samples vs 14?

The conventional 2H blocking uses a fixed window of N_{fix} samples before making a decision (it must wait for the full DFT window to stabilise). The SPRT makes its decision *as soon* as the LLR crosses a boundary — for clear-cut cases (very high or very low ρ_k), the decision comes in 2–3 samples; the average across all cases is 7. The 14-sample average for 2H blocking reflects the fixed window that must be waited out even when the evidence is unambiguous.

Complete 87T-SPRT Algorithm

87T-SPRT Decision Procedure (Algorithm from Chapter 3)

1. Compute $I_{\text{diff}}^T, I_{\text{bias}}^T$ from compensated CT phasors
2. Compute adaptive bias slope: $k_T^*(k_{\text{ibr}})$
3. If $I_{\text{diff}}^T > k_T^* \cdot I_{\text{bias}}^T + I_0^T$ (bias threshold exceeded):
 - (a) Initialise: $\Lambda \leftarrow 0, k \leftarrow 0$
 - (b) **While** $B < \Lambda < A$:
 - i. Compute $\rho_k = I_{2f}(t_k)/I_{1f}(t_k)$ from current DFT window
 - ii. Update: $\Lambda \leftarrow \Lambda + \ln(f_1(\rho_k)/f_0(\rho_k))$
 - iii. $k \leftarrow k + 1$
 - (c) If $\Lambda \geq A$: **Trip** (internal fault confirmed)
 - (d) Else: **Restrain** (inrush confirmed)
4. Else: **Restrain** (below bias threshold)

Adaptive 87T Bias Slope $k_T^*(k_{\text{ibr}})$

The SPRT replaces the inrush discrimination step. The bias slope that governs the *sensitivity* of the differential relay is separately adapted:

$$k_T^*(k_{\text{ibr}}) = 2\varepsilon_{\text{CT}} + \frac{k_{\text{ibr}} \cdot I_{\text{rated}}^T}{I_{\text{bias}}^T} \cdot \frac{\sin(\varphi_B - \varphi_A)}{1 + k_{\text{ibr}}}$$

When $k_{\text{ibr}} \rightarrow 0$: $k_T^* \rightarrow 2\varepsilon_{\text{CT}}$ (classical CT-error-only setting). When $k_{\text{ibr}} > 0$, the second term accounts for the angular ambiguity introduced by the IBR current limiter firmware. This is the same structural derivation as the adaptive 87L bias slope of Chapter 3.

How to Read the Results Table

SPRT vs 2H Blocking: Metric-by-Metric

Inrush rejection (94.1% $\rightarrow$ 99.6%): The conventional 2H blocker correctly restrains 94.1% of inrush events but misses 5.9% (false trips during inrush). The SPRT accumulates evidence over multiple samples — a transiently high ρ_k that quickly returns to baseline is quickly classified as inrush, whereas the 2H blocker only tests the instantaneous ratio. The 99.6% SPRT inrush rejection matches the Wald $\beta = 0.01$ guarantee.

Internal fault TPR (96.8% → 99.1%): The conventional 2H blocker misses 3.2% of internal faults because IBR harmonic injection raises I_{2f}/I_{1f} above h_{th} , triggering the block even during a fault. The SPRT's H_1 likelihood model accounts for IBR harmonic contamination through the adaptive $\mu_1(k_{ibr})$ update, and the accumulation of evidence over 7 samples makes the test insensitive to transient IBR harmonic spikes.

IBR harmonic FP (8.7% → 0.9%): This is the headline improvement. An IBR harmonic false positive occurs when the relay blocks a genuine fault because the 2nd-harmonic ratio is elevated by IBR injection. The 90% relative reduction (from 8.7% to 0.9%) is the direct consequence of the adaptive H_1 model — the SPRT “knows” that some 2nd-harmonic content is expected even during IBR faults and adjusts its discrimination boundary accordingly.

Mean decision samples (14 → 7): The 2H blocker waits for a full DFT window (14 samples at the test rate) before making a decision. The SPRT decides in as few as 2–3 samples for clear cases, with the average at 7 samples. This halving of decision time corresponds to halving the blocking period for inrush detection and halving the delay for fault confirmation.

Best Supporting Figure for This Slide

RECOMMENDED: TR-03/2026 Table 3 (Page 12) — 87T Event Classification

What it shows: The complete 87T event classification table for all 7 canonical scenarios: normal load (no trip), energisation inrush (blocked by $k_2 = 0.42$), overexcitation (blocked by k_5), external fault with CT saturation (model-vetoed, $f_{int} = 0.45$, $\kappa_n = 4.9$), and three internal fault types (all correctly tripped). The table shows the conventional relay decision, final decision, decision source, condition number κ_n , confidence score c_{conf} , internal fault indicator f_{int} , and 2nd-harmonic ratio k_2 for each event.

Why it is the best available: While a dedicated SPRT trajectory figure (showing Λ_k accumulating to cross boundary A or B) would be ideal, the thesis presents the SPRT framework with a complete algorithm and validated results that confirm the key claim: the 5-parameter zone model correctly handles all 7 canonical transformer events. The table directly illustrates how the three blocking mechanisms (inrush: k_2 , overexcitation: k_5 , CT saturation: f_{int} via Stage-2 veto) work together.

Source: TR-03/2026, page 12, Table 3. In PDF_COLLECTION/01_SAMBP_Reports/TR-03_sambp_report.pdf.

Secondary figures / materials to prepare:

- **Draw a SPRT walk diagram:** On a whiteboard or slide, show Λ_k starting at 0, accumulating upward for a fault (crosses $A = +4.6$ in 7 samples) and downward for inrush (crosses $B = -4.6$ in 7 samples). This is the most impactful visual for this slide.
- **TR-03 Figure (87T characteristic):** The dual-slope I_{diff} vs I_{bias} operating plane with 7 events plotted — currently a placeholder but the points (blocked, tripped,

model-vetoed) can be sketched on the plot.

- **TR-03 Table 5** (condition number comparison across 87T/87L/OC): Shows $\kappa_n^{87T} = 4-7$, confirming the 5-parameter model is well-identified.

Cross-Thesis Connections

Connection	Explanation
Adaptive 87L (Slide 9, C1)	The adaptive bias slope $k_T^*(k_{ibr})$ for 87T is derived from the same structural formula as the adaptive 87L slope. Both use k_{ibr} from the EKF digital twin (Chapter 6). The SPRT replaces harmonic blocking in 87T exactly as the LM-adaptive threshold replaces the fixed bias in 87L — parallel adaptive structures for the two differential protection functions.
EKF Digital Twin (Slide 13, C3)	The IBR-adaptive H_1 likelihood model uses $\hat{k}_{ibr}$ from the EKF to compute $\mu_1(k_{ibr})$ — the expected 2nd-harmonic ratio in the fault differential current as a function of IBR penetration. Without the EKF, the H_1 model would use a fixed $\mu_1 = 0.06$ regardless of fleet composition, degrading SPRT performance for high- k_{ibr} scenarios.
LM Inverse Estimation (TR-03, C1)	The SPRT replaces the <i>blocking decision step</i> of the SAMBP Stage-2 gate. The LM inverse estimator still runs to compute k_2 , k_5 , ε_{CT} and the f_{int} indicator; the SPRT then uses the computed $\rho_k = I_{2f}/I_{1f}$ as its observable rather than comparing k_2 against a fixed threshold.
Wald SPRT Theory (Wald 1947)	The SPRT is optimally efficient: Wald's identity proves that no other sequential test requires fewer samples on average to achieve the same (α, β) error rates. This theoretical guarantee is the protection-engineering justification for replacing ad-hoc 2H blocking with a principled statistical test.
IEC 60255 Compliance	IEC 60255-151 specifies maximum false-trip (α) and miss (β) rates for protection relays. The SPRT boundary equations $A = \ln((1 - \beta)/\alpha)$ and $B = \ln(\beta/(1 - \alpha))$ provide a direct mapping from the IEC specification to the relay algorithm, without the ad-hoc threshold tuning required by conventional relays.

Equations to Know for the TSA Meeting

SPRT Log-Likelihood Ratio

$$\Lambda_k = \sum_{i=1}^k \ln \frac{p(x_i | H_1)}{p(x_i | H_0)}$$

Say: “We accumulate evidence sample by sample. Each new observation either strengthens the fault hypothesis or the inrush hypothesis. We decide as soon as the accumulated evidence is conclusive — no fixed window, no guessing.”

SPRT Boundary Equations (IEC 60255 Connection)

$$A = \ln \frac{1 - \beta}{\alpha}, \quad B = \ln \frac{\beta}{1 - \alpha}$$

For $\alpha = \beta = 0.01$: $A \approx +4.60$, $B \approx -4.60$.

Say: “The boundaries A and B are not tuned by trial and error. They follow directly from the required false-trip rate α and miss rate β specified by IEC 60255. The Wald boundary theorem guarantees that these values achieve exactly those error rates.”

LLR Increment for Gaussian Harmonic Models

$$\ln \frac{f_1(\rho_k)}{f_0(\rho_k)} = \frac{(\rho_k - \mu_0)^2}{2\sigma_0^2} - \frac{(\rho_k - \mu_1)^2}{2\sigma_1^2} + \ln \frac{\sigma_0}{\sigma_1}$$

with $\mu_0 = 0.35$, $\sigma_0 = 0.08$ (inrush), $\mu_1 = 0.06$, $\sigma_1 = 0.03$ (fault).

Say: “This is just the log-ratio of two Gaussian PDFs. If the current observation looks more like an inrush waveform, the ratio is negative and we accumulate evidence for H_0 . If it looks like a fault, the ratio is positive. The accumulation is what makes the test robust to individual noisy samples.”

Wald’s Identity: Expected Decision Time

$$\mathbb{E}[N | H_1] \approx \frac{A}{\mathbb{E}[\ln(f_1/f_0) | H_1]} \approx \frac{4.60}{2.73} \approx 1.7 \text{ cycles}$$

Say: “Wald’s identity tells us the average number of samples needed under the fault hypothesis. The SPRT achieves 7 samples on average — half the 14 samples of fixed-window 2H blocking — while simultaneously providing better error rate guarantees.”

Speaker Script**What to Say at the TSA Meeting**

Opening (15 seconds): “Chapter 6 addresses transformer differential protection. The problem is that Type-4 IBR inverters inject 2nd-harmonic current into the network during normal operation — the same harmonic signature that conventional

relays use to identify inrush. This creates an 8.7% false positive rate: the relay blocks a genuine internal fault because the 2nd-harmonic ratio looks like inrush.”

The solution (20 seconds): “Instead of comparing the 2nd-harmonic ratio against a fixed threshold, we use a Wald Sequential Probability Ratio Test. The SPRT accumulates log-likelihood evidence sample by sample and decides as soon as the evidence crosses a statistical boundary. The boundary thresholds A and B are set directly from the IEC 60255-specified false-trip and miss-rate requirements — no ad-hoc tuning. SAMBP further adapts the H_1 likelihood model in real time using the IBR penetration ratio from the EKF digital twin, shifting the expected fault harmonic level as the fleet composition changes.”

Results (20 seconds): “The results are on the right. Inrush rejection rises from 94.1 to 99.6%. Internal fault TPR rises from 96.8 to 99.1%. But the headline is IBR harmonic false positives: from 8.7% down to 0.9% — a 90% relative reduction. And the mean decision time halves from 14 to 7 samples, because the SPRT decides as soon as the evidence is conclusive rather than waiting for a fixed window.”

Theory link (15 seconds): “Wald proved in 1947 that the SPRT is optimal: no other sequential test requires fewer samples on average to achieve the same error rates. So we are not trading speed for accuracy — we get both improvements simultaneously, which is exactly what protection engineering requires.”

Closing link (10 seconds): “This adaptive transformer differential scheme, together with the adaptive line differential from the previous slide, completes Contribution C1 of the SAMBP framework: physics-constrained adaptive differential protection for both lines and transformers in IBR-dominated networks.”

Anticipated Committee Questions and Answers

Q1: The SPRT assumes the observations are i.i.d. Is that valid for ρ_k ?

Answer: Strictly, ρ_k is computed from overlapping one-cycle DFT windows, which introduces correlation between successive samples. The i.i.d. assumption is an approximation. In practice, the SPRT with correlated observations still provides boundary guarantees that are close to the i.i.d. case when the correlation decays faster than the decision time — which holds here because the harmonic envelope evolves on timescales of several cycles (much longer than the 7-sample decision time). A rigorous treatment would use the SPRT for Markov-dependent observations (Wald–Wolfowitz generalisation); this is identified as a theoretical future work item in the thesis.

Q2: How are the H_1 likelihood parameters validated?

Answer: The parameters are estimated from Monte Carlo simulation of IBR-fed internal transformer faults across the parametric envelope ($k_{\text{ibr}} \in [0.06, 0.85]$, 1000 trials per operating point). The Gaussian fit is validated by a Kolmogorov-Smirnov test on the simulated ρ_k trajectories. The SAMBP adaptive update $\mu_1(k_{\text{ibr}})$ shifts the mean upward for high- k_{ibr} conditions where IBR harmonic injection contami-

rates even the fault current, maintaining discrimination. Experimental validation on physical transformer and IBR hardware is identified as the key future validation step.

Q3: Why not use a Generalised Likelihood Ratio Test (GLRT) instead of SPRT?

Answer: The GLRT estimates unknown parameters by maximum likelihood and computes the likelihood ratio at the MLE. For the 87T application, the unknown parameters (IBR harmonic model parameters) are already estimated by the LM inverse estimator running in parallel. The SPRT reuses these estimates via the adaptive $\mu_1(k_{\text{ibr}})$ update, achieving the parameter adaptation benefit of the GLRT without the additional computational cost of a nested optimisation. The SPRT also provides stronger theoretical guarantees (Wald's optimality theorem) than the GLRT in the sequential setting.

Q4: The 2nd-harmonic blocking achieves 94.1% inrush rejection. That already seems adequate – why improve it?

Answer: 94.1% sounds good but means 5.9 out of every 100 transformer energisations cause a nuisance trip. For a distribution system with 50 transformers energised per year, this is 3 false trips annually — unacceptable for a protection relay. More critically, the 8.7% IBR harmonic false positive rate means that in an IBR-dominated feeder, approximately 1 in 12 internal fault events is missed by the conventional relay. A missed internal fault allows the fault to persist until backup protection operates, typically causing significantly more equipment damage and longer outage duration.

Q5: How does the SPRT handle the transition between inrush and internal fault during energisation of a faulted transformer?

Answer: This is the hardest scenario for any inrush discriminator. During energisation of a faulted transformer, the differential current contains both inrush (large k_2 , high ρ_k) and fault (elevated I_{diff} , some fundamental-frequency content) signatures simultaneously. The SPRT's Λ_k accumulates a mixture of positive (fault) and negative (inrush) increments, which extends the decision time. The unrestrained high-set element ($I_{\text{diff}} > I_{\text{hs}} = 5$ pu) provides a fail-safe: if the fault current is severe, it bypasses all blocking and trips immediately. For lower-level faults during energisation, the SPRT decision time increases (more samples needed) but the Wald boundary guarantees bound the miss probability. TR-03 Table 3 shows the model-veto gate correctly handles the external-fault-plus-CT-saturation scenario ($f_{\text{int}} = 0.45$, $\kappa_n = 4.9$); the energisation-plus-fault case is a known limitation flagged for future work.

Source documents: TR-03/2026 (SAMBP 87T/87L framework, 5-parameter zone model, 7-event classification), Chapter 3 thesis text (SPRT theory, Wald boundary equations, LLR increment, adaptive bias slope k_T^*).

Thesis location: Chapter 3 (Differential Protection), Contribution C1, Section on 87T-SPRT.

Best figure: TR-03/2026, page 12, Table 3 (87T batch study results, all 7 events correctly classified). Draw SPRT walk diagram (Λ_k trajectory) on whiteboard for maximum visual impact.